Clinton Enwerem

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Dear Technical Recruiter,

Application: Robotics Research Engineer

I implement Python/C++ robotics software, including ROS 2 hardware interfaces, robot descriptions, controllers, motion planners, simulations, and automated tests, then validate the software on physical robots. I am an Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026, with five years of work in dexterous manipulation, motion planning, feedback control, and robot learning.

realhand_l6_ros2 implements a C++ SocketCAN hardware interface and exports joint and tactile state. The package includes contact-gated control, URDF/SRDF, MoveIt 2, mock hardware, virtual-CAN tests, CI, and physical communication and sensing tests. kinematic_planner_ros2 implements RRT*, Informed RRT*, and FCL primitive, mesh, and self-collision queries for N-DOF URDF manipulators. Grasp Distance Fields enforce collision and grasp constraints with a CBF-CLF quadratic program, which solved in 0.09 ms in the reported manipulation experiments.

ParcelStow evaluates imitation policies in simulation. I trained DAgger, Diffusion Policy, and ACT on 297 expert demonstrations of a Unitree G1 and RealHand L6 insertion task. At twice nominal execution speed, expert success was 84% and ACT success was 53%, with 35 of ACT's 47 failures occurring during insertion. The public release contains the repository, data and models, and stage-level outcomes required to reproduce the comparison.

I fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing. Each of three training seeds completed 136/200 tasks versus 129/200 without supervision, recovering seven failed executions without disrupting nominal successes. I developed a visuo-tactile bimanual assembly pipeline with simulation and hardware execution interfaces, combining depth-based pose estimation (RANSAC/DBSCAN, ICP), collision-constrained trajectory optimization (CasADi/IPOPT), and gravity-compensated position control with fingertip-contact monitoring. I evaluated task execution in MuJoCo.

I developed dexreach to connect GPU-accelerated dexterous grasp synthesis and cuRobo motion planning with Isaac Lab sequence validation, evaluating object motion, task-stage completion, and contact-loss stops.

My resume is attached. You can reach me at **enwerem@umd.edu** or **301-416-9430**.

Thank you for considering my application. I am available to start in December 2026. I look forward to discussing how my experience in robot integration, motion planning, and control could support your team's work.

Sincerely,

Clinton Enwerem

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